

HOMOTOPY PULLBACKS

2024-01-09

Reference: [Rijke 2018, Introduction to HoTT, Chapter 10]

Goals

- Define pullbacks using the universal property and characterize the explicit higher data which determines if a commutative square is a pullback.

- Show that pullbacks are uniquely unique

- Given a cospan $f: A \rightarrow X \leftarrow B: g$, define a canonical pullback type $A \times_X B$, show that for any other cone (p, q, H) with vertex C , there is a map

$$\text{gap}(p, q, H): C \rightarrow A \times_X B$$

such that

$$(p, q, H) \text{ is a pullback} \iff \text{gap}(p, q, H) \text{ is an equiv.}$$

- Some families of examples:

- Products

- Fibers

- Fiber products

- Fiber sequences

- Some extra applications:

- Characterization of fiberwise equivalences

- Preservation of truncation and equivalences

- Pullback pasting

- Disjointness of coproducts

Pullback Squares

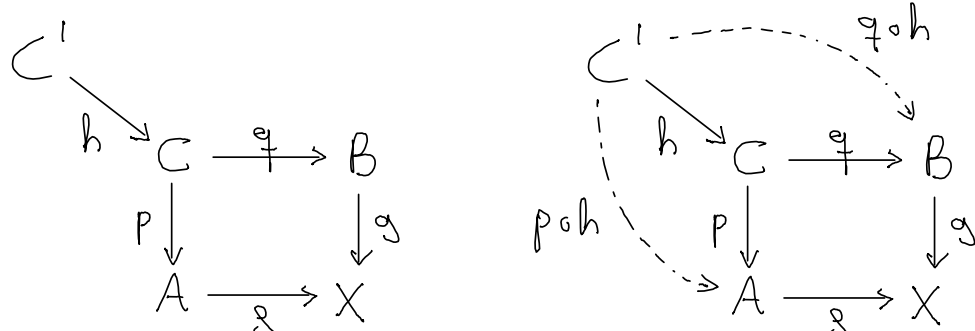
A square
$$\begin{array}{ccc} C & \xrightarrow{q} & B \\ p \downarrow & & \downarrow g \\ A & \xrightarrow{s} & X \end{array}$$
 is said to **commute** if there is a homotopy $H: f \circ p \sim g \circ q$ ($\equiv \prod_{c:C} f \circ p(c) =_X g \circ q(c)$)

$$\left\{ \begin{array}{ccc} \text{Commutative square} \\ \begin{array}{ccc} C & \xrightarrow{q} & B \\ p \downarrow & H \rightrightarrows & \downarrow g \\ A & \xrightarrow{s} & X \end{array} \end{array} \right\} \equiv \left\{ \begin{array}{ccc} \text{Cone over a cospan} \\ \begin{array}{ccc} A & \xrightarrow{s} & X \xleftarrow{g} B \\ \text{with vertex } C \end{array} \end{array} \right\} \equiv \text{cone}(C)$$

where $\text{cone}(C) := \sum_{p:C \rightarrow A} \sum_{q:C \rightarrow B} f \circ p \sim g \circ q$

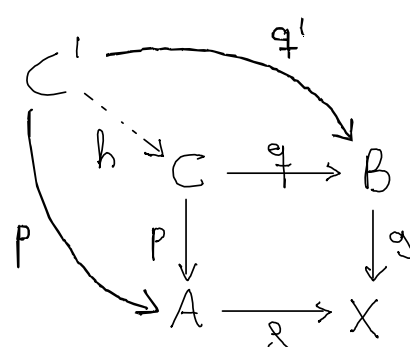
For any commuting square as before and any type C' , define:

$$\begin{aligned} \text{coneMap}(p, q, H): (C' \rightarrow C) &\longrightarrow \text{cone}(C') \\ h &\longmapsto (p \circ h, q \circ h, H \circ h) \end{aligned}$$



A commuting square as before is a **pullback square** if $\text{coneMap}(p, q, H)$ is an equivalence for any type C' .

Corresponds to the universal property of the pullback

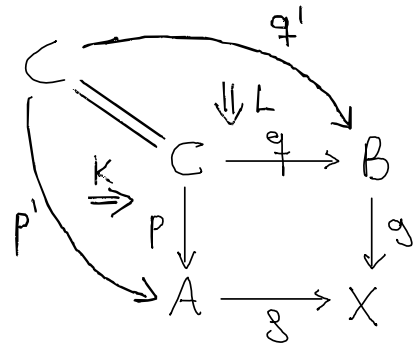


To understand the universal property, first we need to characterize the identity type of $\text{cone}(C)$:

Lemma 1 Let (p, q, H) and (p', q', H') be cones on a cospan $f:A \rightarrow X \leftarrow B:g$ and vertex C . Then

$$((p, q, H) = (p', q', H')) \simeq (K, L, M) \quad \text{where:}$$

$$K: p \sim p', \quad L: q \sim q' \quad \text{and} \quad M: H \cdot (g \circ L) \sim (f \circ K) \cdot H'$$



$$\begin{array}{ccc} f \circ p & \xrightarrow{f \circ K} & f \circ p' \\ H \downarrow & M \rightrightarrows & \downarrow H' \\ g \circ q & \xrightarrow{g \circ L} & g \circ q' \end{array}$$

Proof How we characterize identity type of a type A ?

1. Define a candidate $\prod_{x,y:A} R(x,y)$

2. Prove that R is reflective $\prod_{x:A} R(x,x)$ obtaining a fiberwise map by path induction:

$$\prod_{x,y:A} (x=y) \rightarrow R(x,y) \quad (*)$$

3. Prove that $\sum_{x:A} R(a,x)$ is contractible, and then by the fundamental theorem of identity types, $(*)$ is an equivalence.

In this case,

$$(1) \quad R((p, q, H), (p', q', H')) = (K, L, M)$$

$$(2) \quad (\text{htpyRef}_p, \text{htpyRef}_q, \text{htpyRef}_H): R((p, q, H), (p, q, H))$$

By (3) and associativity of Σ , it suffices to show that

$$\sum_{p'} \sum_{q'} \sum_{H'} \sum_{K:p \sim p'} \sum_{L:q \sim q'} H \cdot (g \circ L) \sim (f \circ K) \cdot H' \simeq$$

is contractible.

Swap the order of q', H', K and $L + \Sigma$ associativity

$$\simeq \sum_{(p', K): \sum_{p'} p \sim p'} \sum_{(q', L): \sum_{q'} q \sim q'} \sum_{H'} H \cdot (g \circ L) \sim (f \circ K) \cdot H' \simeq$$

Both are contractible by function extensionality with centers of contraction $(p, \text{htpyRef}_p)$ and $(q, \text{htpyRef}_q)$

$\Downarrow$ (Exercise: C contractible with center $c \Rightarrow \sum_{x:C} B(x) \simeq B(c)$)

$$\simeq \sum_{H'} H \cdot \text{htpyRef}_{g \circ q} \sim \text{htpyRef}_{f \circ p} \cdot H' \simeq \sum_{H'} H \sim H'$$

Also contractible by fun. ext. $\square$

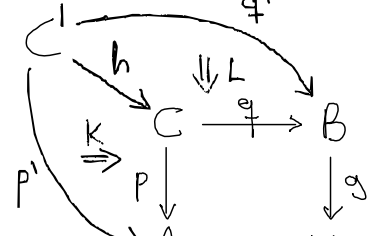
Corollary 2

$$\begin{array}{ccc} C & \xrightarrow{q} & B \\ p \downarrow & H \rightrightarrows & \downarrow g \\ A & \xrightarrow{s} & X \end{array} \text{ is a pullback} \quad \Updownarrow$$

For every type C' and cone (p', q', H') with vertex C' the type of quadruples (h, K, L, M) consisting of

$$\begin{aligned} h: C' \rightarrow C & \quad K: p \circ h \sim p' \\ & \quad L: q \circ h \sim q' \quad M: (H \circ h) \cdot (g \circ L) \sim (f \circ K) \cdot H' \end{aligned}$$

is contractible.



$$\begin{array}{ccc} f \circ p & \xrightarrow{f \circ K} & f \circ p' \\ H \circ h \downarrow & M \rightrightarrows & \downarrow H' \\ g \circ q & \xrightarrow{g \circ L} & g \circ q' \end{array}$$

Uniqueness of pullbacks

The uniqueness of pullbacks up to equivalence follows from:

Lemma 3 Suppose

$$\begin{array}{ccc} C \xrightarrow{q} B & C' \xrightarrow{q'} B & h: C \rightarrow C' \\ p \downarrow H \nearrow \downarrow g & p' \downarrow H' \nearrow \downarrow g & K: p \circ h \sim p' \quad L: q \circ h \sim q' \\ A \xrightarrow{g} X & A \xrightarrow{g} X & M: (H \circ h) \cdot (g \circ L) \sim (f \circ K) \end{array}$$

Then, if any two of the following hold, so does the third:

- (i) C is a pullback
- (ii) C' is a pullback
- (iii) h is an equivalence

Proof By the characterization of the identity types of $\text{cone}(C')$, from (K, L, M) we obtain

$$\text{coneMap}((p, q, H), h) = (p', q', H') \quad (1)$$

For any type D and $k: D \rightarrow C'$ a map,

$$\begin{aligned} \text{coneMap}((p, q, H), (h \circ k)) &\equiv (p \circ h \circ k, q \circ h \circ k, H \circ h \circ k) \\ &\equiv \text{coneMap}(\text{coneMap}((p, q, H), h), k) \\ &\stackrel{(1)}{=} \text{coneMap}((p', q', H'), k) \end{aligned}$$

Thus, there is a commutative triangle:

$$\begin{array}{ccc} (D \rightarrow C') & \xrightarrow{h \circ -} & (D \rightarrow C) \\ \text{coneMap}(p', q', H') \searrow & & \swarrow \text{coneMap}(p, q, H) \\ & \text{core}(D) & \end{array}$$

Then, the result follows from the 3-for-2 property of equiv. and $h \circ -$ is an equivalence for any $D \Leftrightarrow h$ is an equivalence □

As a corollary, we obtain that the pullbacks are not only unique, but "uniquely unique":

Corollary 4 Consider two pullback squares

$$\begin{array}{ccc} C \xrightarrow{q} B & C' \xrightarrow{q'} B \\ p \downarrow H \nearrow \downarrow g & p' \downarrow H' \nearrow \downarrow g \\ A \xrightarrow{g} X & A \xrightarrow{g} X \end{array}$$

Then the type of quadruplet (e, K, L, M) consisting of

$$\begin{array}{l} e: C' \simeq C \quad K: p \circ e \sim p' \quad M: (H \circ e) \cdot (g \circ L) \sim (f \circ K) \cdot H' \\ L: q \circ e \sim q' \end{array}$$

is contractible.

Proof By corollary 2, the type of quadruplet (h, K, L, M) , with $h: C' \rightarrow C$ being a map, is contractible.

By lemma 3, h is an equivalence. Then, because $\text{isEquiv}(h)$ is a non-empty proposition, it is contractible.

The type that we want to study is equivalent to

$$\sum_{(h, K, L, M)} \text{isEquiv}(h) \quad (\text{by } \Sigma\text{-swap})$$

which is a dependent product of a contractible type and a contractible family, thus contractible □

Canonical pullbacks

Let $f: A \rightarrow X \leftarrow B: g$ a cospan. Define the **canonical pullback** $A \times_X B$ of f and g as

$$A \times_X B \equiv \sum_{x:A} \sum_{y:B} f(x) = g(y)$$

$$\pi_1 \equiv \text{pr}_1: A \times_X B \rightarrow A \quad \pi_2 \equiv \text{pr}_1 \circ \text{pr}_2: A \times_X B \rightarrow B$$

$$\pi_3 \equiv \text{pr}_2 \circ \text{pr}_2: f \circ \pi_1 \sim g \circ \pi_2$$

$$\begin{array}{ccc} A \times_X B & \xrightarrow{\pi_2} & B \\ \pi_1 \downarrow & \nearrow \pi_3 & \downarrow g \\ A & \xrightarrow{f} & X \end{array} \text{ is a pullback}$$

Proof $(C \rightarrow A \times_X B) \equiv C \rightarrow \sum_{x:A} \sum_{y:B} f(x) = g(y)$

$$\simeq \sum_{p:C \rightarrow A} \prod_{z:C} \sum_{y:B} f(p(z)) = g(y)$$

$$\simeq \sum_{p:C \rightarrow A} \sum_{q:C \rightarrow B} \prod_{z:C} f(p(z)) = g(q(z))$$

$$\equiv \text{cone}(C)$$

By definition this map is $\text{coneMap}(\pi_1, \pi_2, \pi_3)$.

$$\left(\prod_{x:A} \sum_{y:B(x)} C(x, y) \simeq \sum_{(f: \prod_{(x:A)} B(x))} \prod_{x:A} C(x, f(x)) \right)$$

$$h \longmapsto (\lambda x. \text{pr}_1(h(x)), \lambda x. \text{pr}_2(h(x)))$$

Given a commutative square $\begin{array}{ccc} C & \xrightarrow{q} & B \\ p \downarrow & H \nearrow & \downarrow g \\ A & \xrightarrow{f} & X \end{array}$ define the **gap map**

$$\text{gap}(p, q, H): C \rightarrow A \times_X B$$

$$z \longmapsto (p(z), q(z), H(z))$$

and define $\text{isPullback}(p, q, H) \equiv \text{isEquiv}(\text{gap}(p, q, H))$

Theorem 5 $\begin{array}{ccc} C & \xrightarrow{q} & B \\ p \downarrow & H \nearrow & \downarrow g \\ A & \xrightarrow{f} & X \end{array} \text{ is a pullback} \iff \text{There is a term of } \text{isPullback}(p, q, H)$

Proof By lemma 3, it suffices to show that there exist

$$K: \pi_1 \circ \text{gap}(p, q, H) \sim p \quad L: \pi_2 \circ \text{gap}(p, q, H) \sim q$$

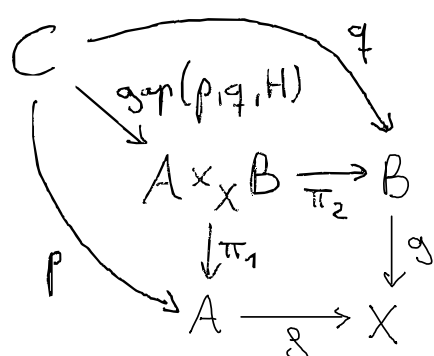
$$K \equiv \lambda z. \text{refl } p(z)$$

$$L \equiv \lambda z. \text{refl } q(z)$$

$$M: (\pi_3 \circ \text{gap}(p, q, H)) \cdot (g \circ L) \sim (f \circ K) \cdot H$$

$$M \equiv \lambda z. \text{rightUnit}(H(z)) \cdot (\text{leftUnit}(H(z)))^{-1}$$

$$M(z): H(z) \cdot \underbrace{(\text{ap}_g(\text{refl } q(z)))}_{\text{refl } g(q(z))} = \underbrace{(\text{ap}_f(\text{refl } p(z)))}_{\text{refl } f(p(z))} \cdot H(z)$$



Fiber products and fibers as pullbacks

Product:

$$\begin{array}{ccc} A \times B & \xrightarrow{pr_2} & B \\ \text{The square } pr_1 \downarrow & & \downarrow \text{const}_* \\ A & \xrightarrow{\text{const}_*} & 1 \end{array} \quad \begin{array}{l} \text{commuting by } \text{const}_{\text{refl}_*} \\ \text{is a pullback} \end{array}$$

Proof By theorem 5, it suffices to show that

$$\text{gap}(pr_1, pr_2, \text{const}_{\text{refl}_*})$$

is an equivalence. The inverse is $\lambda(a, b, p). (a, b)$ □

Pullbacks as fiber products

Theorem 6 Let P and Q be families over X . Then the square

$$\begin{array}{ccc} \sum_{x:X} P(x) \times Q(x) & \longrightarrow & \sum_{x:X} Q(x) \\ \lambda(x, (p, q)). (x, p) \downarrow & & \downarrow pr_1 \\ \sum_{x:X} P(x) & \xrightarrow{pr_1} & X \end{array} \quad \begin{array}{l} \text{commuting by} \\ H \equiv \lambda(x, (p, q)). \text{refl}_x \end{array}$$

is a pullback square.

Proof By theorem 5, it suffices to show that the gap map is an equivalence. The gap map by definition is

$$\lambda(x, (p, q)). ((x, p), (x, q), \text{refl}_x)$$

The inverse of this map is:

$$\lambda((x, p), (y, q), \alpha). (y, (\text{tr}_p(\alpha, p), q))$$
□

The fiber of $f: A \rightarrow B$ at $b: B$ is the type

$$\text{fib}_f(b) \equiv \sum_{a:A} f(a) = b$$

Like the homotopy fiber or the preimage of a function.

For any map $f: A \rightarrow B$, there is an equivalence

$$e: A \simeq \sum_{y:B} \text{fib}_f(y) \quad (*)$$

$$\begin{array}{ccc} A & \xrightarrow{e} & \sum_{y:B} \text{fib}_f(y) \\ \downarrow f & & \downarrow pr_1 \\ B & & B \end{array}$$

and a commutative triangle

Corollary 7 For any cospan $f: A \rightarrow X \leftarrow B: g$, the square

$$\begin{array}{ccc} \sum_{x:X} \text{fib}_f(x) \times \text{fib}_g(x) & \xrightarrow{\beta} & B \\ \alpha \downarrow & & \downarrow g \\ A & \xrightarrow{g} & X \end{array} \quad \text{is a pullback.}$$

where $\alpha \equiv \lambda(x, ((a, p), (b, q))). a$ and $\beta \equiv \lambda(x, ((a, p), (b, q))). b$

Proof By (*), we can expand the square with the following commutative diagram:

$$\begin{array}{ccccc} & & \sum_{x:X} \text{fib}_g(x) & & \\ & \nearrow e \circ \beta & \downarrow e^{-1} & & \\ \sum_{x:X} \text{fib}_f(x) \times \text{fib}_g(x) & \xrightarrow{\beta} & B & & \\ \downarrow \alpha & & \downarrow g & & \\ \sum_{x:X} \text{fib}_f(x) & \xrightarrow{e^{-1}} & A & \xrightarrow{g} & X \\ & \searrow pr_1 & & & \end{array}$$

By theorem 6, the outer square is a pullback. Finally, composing homotopies we obtain the desired result □

Fibers: The square

$$\begin{array}{ccc} \text{fib}_f(b) & \xrightarrow{pr_1} & A \\ \text{const}_* \downarrow & & \downarrow f \\ 1 & \xrightarrow{\text{const}_b} & B \end{array}$$

commuting by $pr_2: \prod_{t: \text{fib}_f(b)} f(pr_1(t)) = b$ is a pullback

Proof Use corollary 7 with $B \equiv 1$, $X \equiv B$, $g \equiv \text{const}_b$,

then we can check that $\sum_{y:B} \text{fib}_f(y) \times \text{fib}_{\text{const}_b}(y) \simeq \text{fib}_f(b)$ □

The previous pullback motivates the definition of fiber sequences:

A fiber sequence $F \hookrightarrow E \twoheadrightarrow B$ consists of three types, called the base space B , the total space E and the fiber F , with base points $x: F$, $y: E$ and $b: B$ and a pullback

$$\begin{array}{ccc} F & \xrightarrow{i} & E \\ \text{const}_* \downarrow & \lrcorner & \downarrow p \\ 1 & \xrightarrow{b} & B \end{array} \quad \text{such that } i(x) = y \text{ and } p(y) = b.$$

Example For any type family B over A and any $a: A$,

$$\begin{array}{ccc} B(a) & \xrightarrow{\text{const}_*} & 1 \\ \lambda y. (a, y) \downarrow & \lrcorner & \downarrow a \\ \sum_{x:A} B(x) & \xrightarrow{pr_1} & A \end{array} \quad \text{is a pullback}$$

(The gap map is homotopic to $\lambda y. ((a, y), \text{refl}_a)$, which is an equivalence between $B(a)$ and $\text{fib}_{pr_1}(a)$.)

If there is a term $b: B(a)$, then we obtain a fiber sequence

$$B(a) \hookrightarrow \sum_{x:A} B(x) \twoheadrightarrow A$$

Applications

Characterization of fiberwise equivalences

Consider a fiberwise map $g: \prod_{x:A} P(x) \rightarrow Q(x)$.

There is a map $\text{total}(g): \sum_{x:A} P(x) \rightarrow \sum_{x:B} Q(x)$
 $(x, p) \mapsto (x, g(x, p))$

such that

g is a fiberwise equivalence $\iff$ $\text{total}(g)$ is an equivalence
 ($g(x)$ equivalence for any $x:A$)

This setting can be further generalised: consider $f:A \rightarrow B$
 and a fiberwise map $g: \prod_{x:A} P(x) \rightarrow Q(f(x))$, then define

$$\text{total}_f(g): \sum_{x:A} P(x) \rightarrow \sum_{y:B} Q(y)$$

$$(x, p) \mapsto (f(x), g(x, p))$$

Then, we can characterise when g is a fiberwise equivalence:

Theorem Let $f:A \rightarrow B$ and $g: \prod_{x:A} P(x) \rightarrow Q(f(x))$. Then

$$g \text{ is a fiberwise equivalence} \iff \begin{array}{ccc} \sum_{x:A} P(x) & \xrightarrow{\text{total}_f(g)} & \sum_{y:B} Q(y) \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \end{array} \text{ is a pullback}$$

Preservation of properties

A map $f:A \rightarrow B$ is **k -truncated** if for each $y:B$ the fiber $\text{fib}_f(y)$ is a k -truncated.

(-2) -truncated maps $\equiv$ fibers contractible $\iff$ Equivalences

(-1) -truncated maps $\equiv$ fibers propositions $\iff$ Embeddings

$\left(f:A \rightarrow B \text{ is an embedding if } \text{ap}_f:(x=y) \rightarrow (f(x)=f(y)) \right)$
 is an equivalence

In general, f is $(k+1)$ -truncated $\iff \text{ap}_f$ is k -truncated

Theorem Consider a pullback square

$$\begin{array}{ccc} C & \xrightarrow{q} & B \\ p \downarrow & \lrcorner & \downarrow g \\ A & \xrightarrow{f} & X \end{array}$$

If g is k -truncated, then so is p

Pullback pasting Consider a commuting diagram

$$\begin{array}{ccccc} A & \xrightarrow{k} & B & \xrightarrow{l} & C \\ f \downarrow & & \downarrow g & & \downarrow h \\ X & \xrightarrow{i} & Y & \xrightarrow{j} & Z \end{array}$$

with homotopies $H:i \circ f \sim g \circ k$, $K:j \circ g \sim h \circ l$ and
 $(j \circ H) \cdot (K \circ k): j \circ i \circ f \sim h \circ l \circ k$

If the right square is a pullback, then

The left square is a pullback $\iff$ The outer square is a pullback

Disjointness of coproducts For any A and B ,

$$\begin{array}{ccc} 0 & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & A+B \end{array} \text{ is a pullback}$$